Adaptive traffic signal optimization considering emergency vehicle preemption and tram priority using PVS algorithm

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ABSTRACT
The tram has been known as a modern, environmentally friendly, quick and safe means of transportation; hence, giving the signal priority for trams on arterial streets at signalized intersections is crucial to improve tram service efficiency. However, tram priority can impact the movement of emergency vehicles (ambulances, police cars, fire engines, etc) which required quick response to save lives and properties. Therefore, to organize the traffic road, it is necessary to make the urban road network more efficient by considering both tram and emergency vehicles priority. In this paper we propose a new approach, a system that can adapt efficiently with a priority signal control. Thereof, we have used Transit Signal Priority (TSP) techniques to adjust signal phasing in order to serve emergency vehicles and manage the tram priority in a coordinated tram intersection. The system used in this study provides acyclic signal operation based on a real time control approach including: - An optimization process using passing vehicle search algorithm (PVS) to control the flow through the network. - A database to manage the sensor data from detectors for real time predictions of emergency vehicle and tram arrival time. - A traffic road simulation module to evaluate best timing plans at the intersection.

CCS Concepts
Theory of computation → Sequential decision making; Applied computing → Operations research → Transportation

Keywords
Tram priority; emergency vehicle preemption; Transit Signal Priority techniques; coordinated tram intersection; optimization; Passing Vehicle Search algorithm; traffic road simulation

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1. INTRODUCTION
During the last years, Morocco has known a significant progress of urban areas. The flows of travels invade very widely the public place, whose structure and shape are more and more dependent on motorized travels, and they encourage the explosion of the car traffic. Indeed, the Moroccan cities undergo enormous pressures due to their incapacity to supply a road traffic management that is adapted to the needs of the population. Congestion of the traffic, insecurity and pollution degrade more and more the living conditions in towns, that's why the dissatisfaction of the users has increased towards the dysfunctions observed, during the vastest periods of the day. The conventional method adopted to prevent or reduce congestion in modern cities is based on road signs (the use of standard color lights). In Morocco, the current traffic lights system is isolated. The used programming, is summarized in two (exceptionally three) successive phases, often in time fixed and with a single program whatever is the request without regard for the pedestrians. To overcome these shortcomings, several relevant experiences for arrangement of road lighting were designed by Moroccan cities. In the proposed system, we aim at optimizing and regulating the real time traffic with equipments that have an adaptive programming which supply average innovators for the traffic management [1][2]. This system will be essentially based on the collection of information from detectors installed on the axes of traffic and intersections, to preempt arrival time for the oncoming emergency vehicles and manage the tram priority at coordinated tram intersection. The sensor data from detectors will be stocked in a database and then be used with TSP techniques to adjust signal phasing (phases sequence and green duration) through delay optimization. Thus, to evaluate best timing plans at the intersection, a traffic road simulation module is developed. Several mathematical models were developed in the literature for real time signal timing optimization [3]. Most of these approaches lean on criteria to minimize vehicles delay and number of stops. SCATS [4], OPAC [5] and SCOOT [6] used a performance index based on the average queue length and the number of stops observed at all approaches in the network. Furthermore, delay minimization was also used by Gazis [7], Gazis and Potts [8], D'Ans and Gazis [9] Michalopoulos and Stephanopoulos [10], Zhang et al. [11] Lo and Chow [12] Gartner et al. [5] Lucas et al. [13] Mirchandani and Head [14] and Van Katwijk [15] Considerable works has already been carried out in the minimization of the general traffic delay associated with emergency vehicle preemption [16] [17] and

several mathematical models were developed in the literature for transit signal priority optimization [18] [19]. This paper is organized as follows: section 2 presents related works, section 3 describes the systems and methods of optimization used in our case study; we present in section 4 our methodology and in section 5 our test results, we draw finally the conclusion of this paper in section 6.

2. Related works

Genetic algorithm was widely used to solve the problem of signal timing optimization. The first study appears in [20] after that, to optimize the four basic signal timings Hadi and Wallace [21] combined GAs with TRANSYT-7F. In the same context Park et al. [22] optimize all four basic signal timing parameters simultaneously with GA. Their work was further extended to consider multiple optimization strategies in Park et al. [23]. Abu-Lebdeh and Benekohal [24] used micro-genetic characteristics such as convergence, population size and performance [25] to solve the problem of signal control optimization in a dynamic environment. GA was used as an optimizer and VISSIM as a micro simulation tool by Park and Schneeberger [26] to evaluate the performance of five different signal timing plans on a real-world arterial. Genetic algorithm was used in [27] to optimize the regulation of timing plan of nine intersections situated in the city of Chicago United States. GA optimization was also used to consider the benefits of optimizing controller, detector, and volume-density related settings [28]. Turky et al. developed a GA-based algorithm that dynamically optimizes splits to control the flow of both vehicles and pedestrians at an intersection [29]. The performance comparison showed that the GA-based algorithm outperformed a fixed-time controller. Singh et al. integrated a GA to provide optimal green time extensions for a set of signals in real time, and then used a traffic emulator to model dynamic traffic conditions [30]. A GA combined with Hill Climbing to optimize and adjust signal timings was presented in [31].

The experimental results confirmed the efficiency of a real time based control system compared to a fixed time based traffic system.

In [32], the authors studied the behavior of the driver during the changes of traffic lights. Their approach was based on a genetic algorithm and it was tested in the city of Chester, UK. The genetic algorithm was also used in [33] to optimize the cycles of the three-colored traffic lights in a shopping district in the city of Santa Cruz, in Spain. A new hybrid algorithm based on NSGA-II and local search is presented [34] to solve mutiobjective signal timing problem, by optimizing two objectives which are, the minimization of total time lost in the traffic and the maximization of throughput. The simulation was realized with Sumo micro simulation. In most of these studies, GAs were better at generating timing plans than other optimization algorithms.

Several works were led to solve the problem of signal timing regulation by using particle swarm optimization to find the optimal duration of the cycle. In [35], PSO was used as an optimizer which collaborates with a controller, designed with logic floue, installed in each intersection of the network. Indeed, PSO was used to define the rules of the controller, to detect the optimal duration of the green time for every phase of traffic lights.

A multi-objective algorithm based on PSO using a strategy of control based on a predictive model to optimize the duration of the cycle, was presented in [36], the proposed algorithm was tested in a city network consisted of 16 intersections and 51 ways.

In [37], PSO was also used to maximize throughput and minimize total trip of vehicles. The evaluation of the cycle was based on the microscopic simulator SUMO. The proposed algorithm was tested in an urban area situated in the city of Malaga and Sevilla in Spain and in Bahia Blanca, in Argentina.

More recently, PSO is used in particular to minimize the duration of green time and fuel consumption [38], [39].

In [40] the authors presented a new model based on PSO to minimize the average delay in a traffic circle, combined with VISSIM microsimulation, this model was developed in 28 signalized intersections in the city of Izmir in Turkey. In the same context Sina Dabiri and Montasir Abbas [41] implemented PSO with VISSIM to optimize the performance index in the intersections on Prices Fork Road, Blacksburg in Virginia. The method proposed, gave good results with regard to the results obtained in diverse situations with the Vistro optimizer based on the genetic algorithm.

3. Traffic signal timing optimization

3.1 Transit Signal Priority

Transit signal priority (TSP) is an operational strategy that facilitates the movement of transit vehicles in service like buses or emergency vehicles which required quick response to improve efficiency at signalized intersections. There are three categories of TSP priority strategies: - Passive signal priority which is the initial form of TSP, it's independent from the real time traffic information. - Active priority provides priority treatment to a specific transit vehicle following two strategies when a TSP-equipped vehicle is detected. The first one is the green extension strategy which extends the green time for the TSP movement and the second one is the early green strategy which shortens the green time of preceding phases to expedite the return to green for the movement. - Real time or adaptive optimization minimizes weighted vehicle delay or passengers and optimizes signal plan considering bus dwell time at stops. Similarly, inductive loop detectors are installed under the tramway to give priority to the tram; the demand detector is situated at between 150m and 200m from the junction, as a tram passes over it a demand is registered with the signal controller to move as quickly as possible to a point in the signal sequence where the tram receives a signal to proceed. A second detector is situated at the tram stop line, as the tram passes over it the priority demand is removed and the signals will move off the tram signal. A third detector is situated on the exit side of the junction, as the tram passes over it the signals drop out of tram mode (LRT) and move to serve the next phase around the junction.

Tram Travel Time TTT is calculated as follows:

$$TTT = (detector\ distance)\ /\ (Average\ free\ speed) \quad (1)$$

3.2 Early green and green extension strategy

3.2.1 Emergency vehicle priority Algorithm

Video sequence of road traffic can be analyzed to detect emergency vehicles entering the phase group and then get their positions and speeds, the emergency vehicles are ambulances, fire engines and police cars.

Algorithm 1.1
1. Video sensor detects the presence of EV.
2. Define the phase group.

3. Determine emergency vehicle position and speed.

4. Calculate the distance between the emergency vehicle and the intersection (Euclidean Distance)

5. The controller checks the arriving time emergency vehicle.

6. Generate the timing plan (optimization process) based on the measured distance values and send these values to the traffic lights.

7. Ensure that the phase group is in service at its estimated arrival time,

8. ***if*** *phase group is already green* ***then***

9. apply Green extension strategy,

10. ***else*** *apply early green strategy.*

11. Return phase group sequence.

12. ***end if***

3.2.2 Tram priority Algorithm

Algorithm 1.2

1. As a tram passes over inductive loop (situated between 150m and 200m from the junction) a demand is registered with the signal controller to move as quickly as possible to a point in the signal sequence where the tram receives a signal to proceed.

2. Calculate Tram Travel Time (eq. 1)

3. Determine the green light duration based on Tram Travel Time for the current phase.

4. Ensure that the tram phase is in service at its estimated arrival time,

5. ***if*** *Tram phase group is green* ***then***

6. Apply Green extension strategy,

7. ***Else*** *apply early green strategy.*

8. As a tram passes over inductive loop situated on the exit side of the junction, the signals drop out of tram mode (LRT) and move to serve the next phase around the junction.

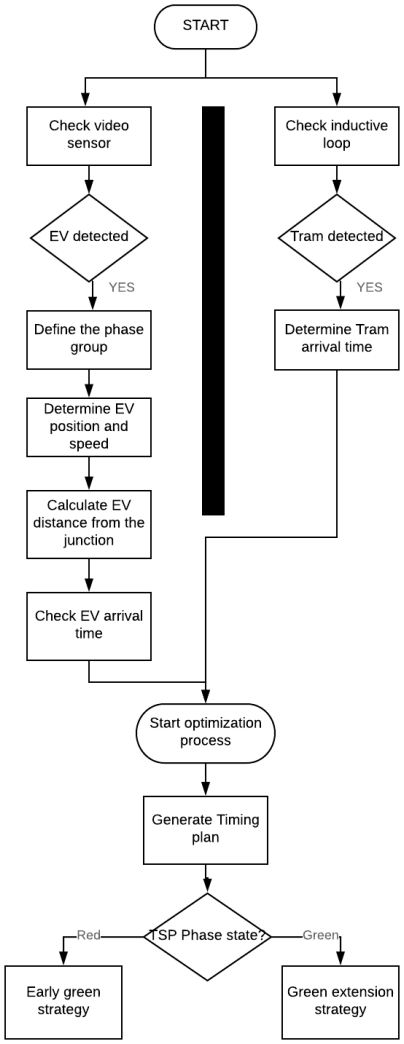


Figure 1. Early green and green extension strategy

3.3 Vehicle Delay Minimisation

$\forall h \in N_h$

Let D be the objective function to minimize total vehicle delay in the intersection [42]

$$D^h = \sum_i^{N_i} \sum_j^{N_j^i} \sum_x^{X_{ij}} d_{ij}^h(x)\{s_i^h(k), p_i^h(k)\} \quad (2)$$

Subject to the following constraints

$for\ \forall i \in N_i, j \in N_j^i$ And $\forall h \in N_h$

$$h_L = \sum_k^{K_i^h} p_i^h(k)\ \ for\ k = 1, \dots, \mathrm{K_i^h} \quad (3)$$

$$h_L \geq p_i^h(k) for\ k = 1, \dots, K_i^h \quad (4)$$

$$g_{min} \leq \begin{cases} p_i^h(k) + p_i^{h-1}\left(\theta_i^{h-1}\right) if\ k = 1\ and\ s_i^h(k) = s_i^{h-1}\left(\theta_i^{h-1}\right) \\ p_i^h(k)\ else\ if\ k = 2, \dots, K_i^h - 1 \end{cases} \quad (5)$$

Where

h = time horizon

i = intersection

j = approach link

x = vehicle

D^h = total delay during h time horizon

N_i = number of intersection

N_j^i = number of approach links at intersection i

X_{ij} = number of vehicles on approach j at intersection i

N_h = number of time horizons

$d_{ij}^h(x)$ = delay of vehicle x on approach j at intersection i during h time horizon

$s_i^h(k)$ = sequence of kth phase at intersection i during h time horizon

$p_i^h(k)$ = duration of kth phase at intersection i during h time horizon

h_L = length of time horizon

g_{min} = minimum phase length

θ_i^h = number of phases in implemented fraction timing at intersection i during h time horizon

K_i^h = number of phases at intersection i during h time horizon

4. Methodology of our architecture

The architecture of our system includes three parts, Passing vehicle search optimizer, database connection and the simulator integration. Through this section, we broadly describe the methodology of our architecture.

4.1 Passing vehicle search algorithm for signal timing optimization

A meta-heuristic is a general framework for heuristics for solving hard global optimization problems, most meta-heuristics are based on some kind of natural phenomenon. Passing Vehicle Search algorithm [43], similar to other meta-heuristics is also a population based method which requires an initial set of solutions to start and it then searches for the optimum solution by following the mathematical characteristics of vehicles overtaking on a two-lane highway.

Algorithm 1.3

Step 1: *Define the parameters such as population size (PS), stopping criteria (Number Of Generations (NOG), Maximum function evaluations (FE), and error) number of design variables (DV), bounds on design variables ((LB),(UB)).*

Define the function for optimization:

Minimize f(x)

subject to $X=\{X_1, X_2, \dots, X_{DV}\}$, *where* $LB_i \le X_i \le UB_i$

Step 2: *Generate an initial set of solutions.*

Step 3: *Select any three set of parameters from the given solution space and out of which one solution represents the Current/Back Vehicle (BV) and the other two solutions are selected randomly and represent the Front Vehicle (FV) and the oncoming Vehicle (OV).*

Step 4: *Evaluate the Distance (D) and Velocities (V) for the three set of parameters (vehicles).*

The distances between the vehicles are calculated based on the fitness value. Arrange the population in ascending order and calculate the distance using Eq. 6. From the expressions, it can be observed that the value of the distance is normalized to 1. The velocity corresponding to the vehicle is calculated using Eq. 7

$$D_k = \frac{N_k}{PS} \qquad (6)$$

$$V_k = R_k(1 - D_k) \qquad (7)$$

where k=(1,2,3) and R_k *is a random number* $\in(0,1)$

Step 5: *Depending upon the velocity and distance of the set of parameters, update the solution space and if the solution is better than previous solution, accept the new solution (new parameters).*

Step 6: *Repeat the procedure until the termination criteria is satisfied.*

4.2 Description of our system

In our approach, we distinguish vehicle detection, emergency vehicle detection and tram detection.

- For emergency vehicle (EV), GPS technology is the first helps to limit the coordinates of our crossroads, so if an EV coordinates are detected then it's up to video sensor to detect the vehicle identification number and calculate its position, Speed and the distance between the emergency vehicle and the intersection to predict the arrival time.

- For the tram, inductive loop situated between 150m and 200m from the junction allow us to predict the tram arrival time, in order to give our model a realistic approach we consider that the speed of tramways upcoming the crossroad is the same. But its speed before arriving to the 150 m loop can be different. Also we consider the number of stops and the frequency of Tramway passage in an hour and we adapted it to our time horizon.

- For vehicle detection, the camera sensor captures the real-time traffic video based on Kalman filter detection in time horizon adopted in the video traffic detection process, when a vehicle passes at the studied area; vehicle identification number, position and speed, traffic density and vehicle count are assigned and then stored in the database.

4.2.1 Priority rules

- If a tram or EV arrives at a traffic signal it will get priority through the junction. The controller will then serve the other demands (depending on vehicle delay minimization for the rest of phases), and also depending on which phase the EV is detected to choose the appropriate strategy.

Strategy 1: Early Green Strategy.

Strategy 2: Green Extension Strategy.

- Both Tram and EV arrive at the junction at the same time, this will also trigger priority through the junction. And the priority always goes to the EV. That's why we proceed to determine a case

series for green duration and phases sequence to optimize traffic signal timing.

- In case neither Tramway nor EV is detected the sequence of phases applied takes in consideration the real time traffic. As a special case if a phase contains no vehicle, the simulation program will ignore it, and as long as there is no vehicle waiting for the "Green light". Phase will be "Red" all the time.

4.3 Traffic simulation module

Simulation is a powerful technique for understanding real time situation. Traffic simulation is used to study traffic congestion at a road. Many studies have been carried out for modeling traffic congestion using simulation models. In our contribution, we propose a traffic simulation module to evaluate timing plans generated by our PVS optimizer. This simulation module provides graphical user interface that can be used for traffic modeling, in which GUI inputs are:

Simulation parameters: - Duration of simulation time (number of time horizons). - Time step during the simulation: In each time horizon, we fix the number of simulations that the program should make, so in order to make the algorithm more performing for a Time Horizon = 60 s , we choose 5s Step .

Database output: - Time : The detection time in the specific time horizon.- Position, - Speed, - link, - vehicle types (Car, Tram, Emergency vehicle), - Sequence of phases, - The database saves all the data of the crossroads, including the number and the passage time of EV / Tramway / ordinary vehicles and the sequence of phases that were applied during the operational time.

Detectors at intersections: The detectors are the main key of our simulation because all the simulation is based on the precision of the data provided by detectors.

Our traffic simulator estimates for a giving timing plan (phase sequence, green duration): - Total vehicle delay on approach J at intersection I. - Number of vehicles on approach J at intersection I. - Number of vehicles exiting approaches J at intersection I. - Average delay per phase. - Average delay per time horizon. All this information is shown in our dashboard in order to give a summary of what is happening in our crossroads. Finally, at each time step the real-time simulation module defines: - State light phase (red, green). - Duration of each phase.

4.4 Case study of tram priority and emergency preemption in Real-Time optimization

In this case study we take as an example the intersection between France and Fal Ouled Omair streets Rabat Morocco, taking into account the priority of the tram [44].

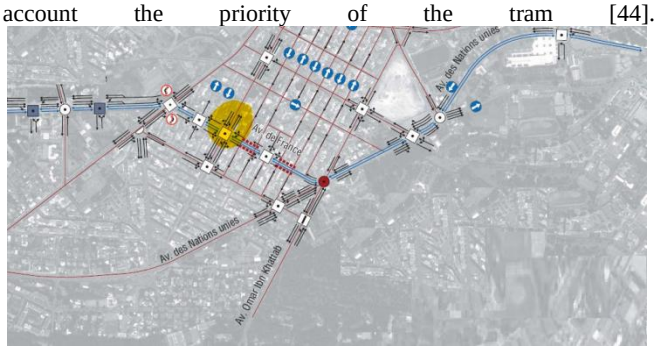


Figure 2. The intersection between France and Fal Ouled Omair streets

1: The database sends all the necessary information to the PVS.

2: The optimization process returns the phase sequence to the Database.

3: The crossroads system gets the phase sequence from the DB and applies it to the crossroads.

4: The Simulator gets all the inputs to display a graphic simulation including all the information of the crossroads (Average delay, etc)

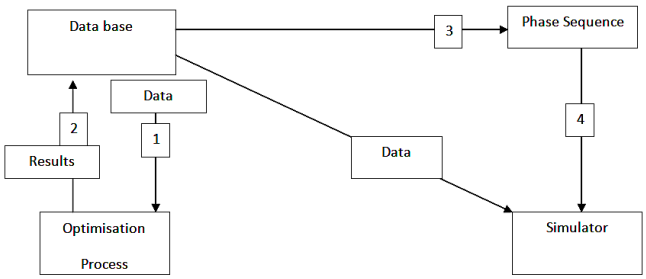


Figure 3. The proposed system

5. Results

The results consider the crossroads with the characteristics below: Length of intersection: 100 m (detection), cruising speed: 5 m/s. Number of vehicles depends on the database, in order to be more precise we used different databases in our simulation. – Rush hour. - Medium Traffic. For each case, we calculated all the parameters and especially in a comparative approach, the total delay in each case.

Table 1. The optimal green light sequence: (Medium Traffic)

Adaptive	Phase 1	Phase 2	Phase 3	Phase 4
Green light	19,0736	19,5289	15,6349	19,5668

Table 2. The static model timing plan: (Medium Traffic)

Static	Phase 1	Phase 2	Phase 3	Phase 4
Green light	15	20	15	20

Total Delay for the Static Model: 3106 s

Total Delay for Adaptive Model: 1592 s

Optimization coefficient = 52%

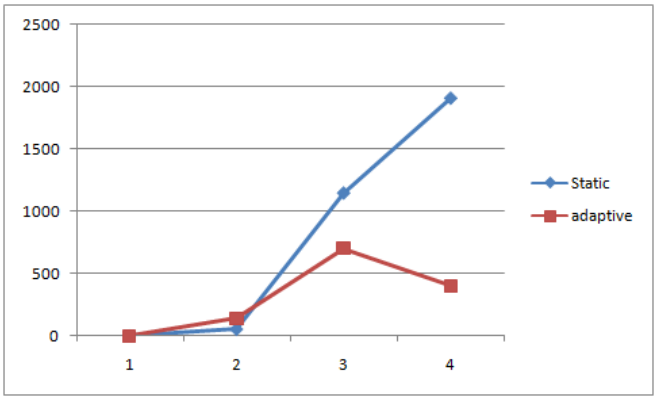


Figure 4. Total delay per phase (Medium Traffic)

Figure 4 describes the minimization of waiting time for a time horizon of 74 seconds and for a total number of 39 of vehicle, equivalent to 2340 vehicle / hour. The coefficient of optimization can go from 0 % to 70 %, because the static model can be optimal in some situations (very low flow of traffic) or when the time of detection of vehicles coincides with the green phase of the intersection.

Table 3. The optimal green light sequence: (Rush hour)

Adaptive	Phase 1	Phase 2	Phase 3	Phase 4
Green light	18	22	17,6	18,5

Total Delay for the Static Model: 6177 s

Total Delay for Adaptive Model: 3042 s

Optimization coefficient = 49%

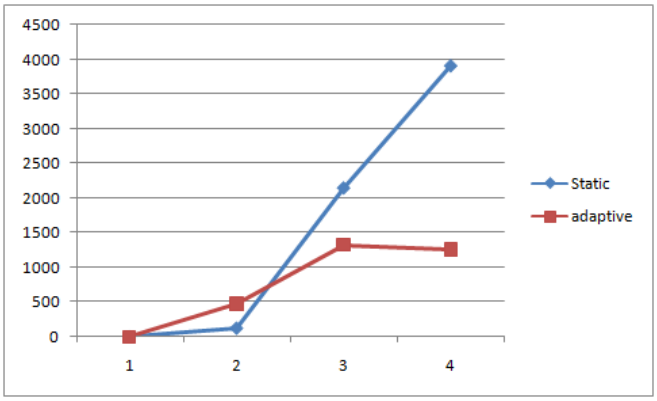


Figure 5. Total delay per phase (rush hour)

Figure 5 presents the waiting time for a time horizon of 76 seconds and for a total number of 55 of vehicle, equivalent to 3300 vehicle / hour.

EV detected in the second phase

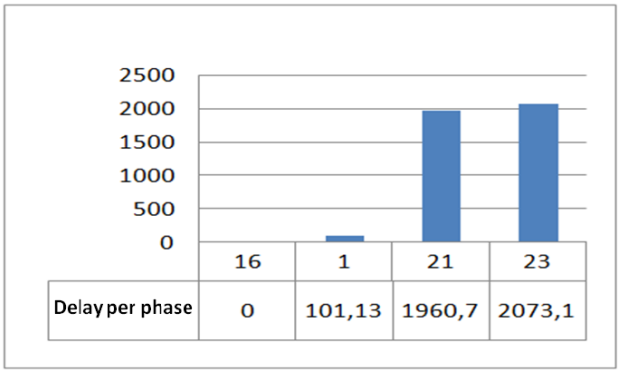


Figure 6. Delay per phase for the adaptive model

In this case (figure 6) the EV was detected in the second phase.

The detection protocol is based on Lambert coordinates. We consider that all the EV are provided with a GPS, so we know exactly the position of each EV in the city. Once close to our crossroads the IHM is aware of it and does the necessary calculations.

Tramway detection:

We mentioned previously when Tramway is detected the phase sequence becomes a list of five values:

- The first 4 values concern the ordinary phase sequences.
- The last value represents the length of the tramway phase.

Table 4. Detection of tram presence

Detection time 300m	Detection time 200m	Speed 200m
35	23.5	8.5

Table 5. Tramway timing plan

Phase 1	Phase 2	Phase 3	Phase 4	Tram
19	19,5	15,6	19,5	12

When tramway is detected the delay of each vehicle increases by the tram phase.

For example, the tram detection time is 35s, which means the second phase is running, as a priority matter the tram will not stop, as a result the third and fourth phase will start with a delay of 12 seconds. In our model all theses parameters are manipulated by the database.

6. Conclusion

The goal of the study is to present an adaptive traffic control at coordinated tram intersections considering emergency vehicle preemption through congested routes, which is the first experience designed for Moroccan road traffic arrangement. The approach combines the measurement of distance between tram and the intersection using inductive loop detectors situated at 200m from the junction and between the emergency vehicle and intersections using GPS technology. The distance between the emergency vehicle and the intersection is calculated using Euclidean distance.

For this end a database is proposed to manage the sensor data from detectors for real time predictions of arrival time flow which will be used as an input for the PVS optimizer that aim to achieve minimal tram and emergency vehicle delay at traffic signals, it returns the phase sequence and green duration for each phase to the Database. Then, a traffic road simulation module is developed to evaluate best timing plans at the intersection. The case study used an intersection with tram priority in Rabat-Morocco and test results prove efficiency of the method.

In future research, the proposed approach should be investigated on a network where multiple emergency vehicles coming on different phases and pedestrian priority will be considered.

7. REFERENCES

[1] Boillot, F., Blosseville, Lesort, Motyka, papageorgiou, sellam, Optimal signal control of urban traffic networks, sixth international conference on road traffic monitoring and control, IEEE london april 1992.

[2] Boumghar, S. (2000). Nouvelle approche de l'optimisation en temps réel des feux d'un carrefour complexe isolé par la programmation linéaire mixte en nombres entiers. Thèse de doctorat, Lamsade, Université Paris Dauphine, Paris, France.

[3] Farges (J.L.), Khoudour (L.), Lesort (J.B.). - PRODYN: On Site Evaluation. IEE Conference, Publication N 320, 1990, pp 62-66.
[4] Lowrie, P. (1982). The Sydney co-ordinated adaptive traffic system: Principles, methodology, and algorithms. In Proceedings of the IEE conference on Road Traffic Signaling, numero 207, pages 67 to 70. IEE Pub.
[5] Gartner, N., F. J. Pooran, and C. Andrews. Implementation of the OPAC Adaptive Control Strategy in a Traffic Signal Network. Proc., Intelligent Transportation Systems, Oakland, Calif., IEEE, New York, 2001, pp. 195–200.
[6] Bretherton, R., Hunt, P., Robertson, D. et Royle, M. (1982). The SCOOT on-line traffic signal optimization technique. In International Conference on Road- Traffic Signalling, London, UK.
[7] Gazis, D. C. Optimum Control of a System of Oversaturated Intersections. Operation Research, Vol. 12, No. 6, 1964, pp. 815–831.
[8] Gazis, D. C., and R. B. Potts. The Oversaturated Intersection. Proc., 2nd International Symposium on the Theory of Road Traffic Flow, Paris, 1965.
[9] D'Ans, G. C., and D. C. Gazis. Optimal Control of Oversaturated Storeand-
Forward Transportation Networks. Transportation Science, Vol. 10, No. 1, 1976, pp. 1–19.
[10] Michalopoulos, P. G., and G. Stephanopoulos. Oversaturated Signal System with Queue Length Constraints I. Transportation Research, Vol. 11, No. 6, 1977, pp. 413–421.
[11] Zhang, L., Y. Yin, and Y. Lou. Robust Signal Timing for Arterials Under Day-to-Day Demand Variations. In Transportation Research Record: Journal of the Transportation Research Board, No. 2192, Transportation Research Board of the National Academies, Washington, D.C., 2010, pp. 156–166.
[12] Lo, H. L., and H. F. Chow. Control Strategies for Oversaturated Traffic. Journal of Transportation Engineering, Vol. 130, No. 4, 2004, pp. 466–478.
[13] Lucas, D. E., P. B. Mirchandani, and K. L. Head. Using Remote Simulation to Evaluate Real-Time Traffic Control Strategies. In Transportation Research Record: Journal of the Transportation Research Board, No. 1727, TRB, National Research Council, Washington, D.C., 2000, pp. 10–15.
[14] Mirchandani, P., and L. Head. A Real-Time Traffic Signal Control System: Architecture, Algorithms, and Analysis. Transportation Research Part C: Emerging Technologies, Vol. 9, No. 6, 2001, pp. 415–432.
[15] Van Katwijk, R. Multi-Agent Look-Ahead Traffic Adaptive Control. PhD dissertation. Delft University of Technology, Netherlands, 2008.
[16] X. Qin and A. M. Khan, "Control strategies of traffic signal timing transition for emergency vehicle preemption," Transportation research part C: emerging technologies, vol. 25, pp. 1–17, 2012.
[17] W. Kang, G. Xiong, Y. Lv, X. Dong, F. Zhu, and Q. Kong, "Traffic signal coordination for emergency vehicles," in 17th International IEEE Conference on Intelligent Transportation Systems (ITSC). IEEE, 2014, pp. 157–161.
[18] Brenner, M. F. (2007). "LRT-and tramway priority in the course of "Green Waves" a planning method to integrate public transport into progressive signal systems." Proceedings of the 14th World Congress on Intelligent Transport Systems, Beijng.
[19] Min Yang, Jian Ding, Wei Wang, and Yi-Yuan Ma. A Coordinated Signal Priority Strategy for Modern Trams on Arterial Streets by Predicting the Tram Dwell Time. KSCE Journal of Civil Engineering (0000) 00(0):1-14
[20]Foy, M. D., R.F. Benekohal, and D.E. Goldberg. "Signal Timing Determination Using Genetic Algorithms", Transportation Research Record 1365, TRB, Washington D.C., 1992, 108-115
[21] Hadi, M.A., and C. E. Wallace. "Hybrid Genetic Algorithm to Optimize Signal Phasing and Timing", Transportation Research Record 1421, TRB, Washington D.C., 1993, 104-112.
[22] Park, B.B., C.J. Messer, and T. Urbanik II. "Traffic Signal Optimization Program for Oversaturated Conditions - Genetic Algorithm Approach", Transportation Research Record 1683, TRB, Washington D.C., 1999, 133-142.
[23] Park, B.B., Rouphail, N.M., and J. Sacks. Assessment of a Stochastic Signal Optimization Method Using microsimulation. Technical Report No 110, National Institute of Statistical Sciences.
[24] Abu-Lebdeh, G., and R. F. Benekohal. "Development of a Traffic Control and Queue Management Procedure for Oversaturated Arterials", Transportation Research Record 1603, TRB, Washington D.C., 1997, 119-127.
[25] Abu-Lebdeh, G., and R. F. Benekohal. "Convergence Variability and Population Sizing in Micro-Genetic Algorithms", Computer-Aided Civil and Infrastructure Engineering, 14 (1999), 321-334.
[26] Park, B., and J.D. Schneeberger. Evaluation of Traffic Signal Timing Optimization Methods Using a Stochastic and Microscopic Simulation Program. Research Report No UVACTS-5-0-4, University of Virginia, Charlottesville, Virginia, January 2003.
[27] N. M. Rouphail, B. B. Park, and J. Sacks, Direct signal timing optimization: strategy development and results. technical report, in XI Pan American conference in Traffic and Transportation Engineering, 2000.
[28] Park, B. B., and I. Yun. Stochastic Optimization Method for Coordinated Actuated Signal Systems. UVACTS-15-0-102, University of Virginia, Charlottesville, Virginia, January 2006.
[29] A.M. Turky, M.S. Ahmad, M.Z.M. Yusoff, and B.T. Hammad , "Using Genetic Algorithm for Traffic Light Control System with a Pedestrian Crossing", In Proceeding of the 4th International Conference on Rough Sets and Knowledge Technology, Gold Coast, Australia, July 14-16, 2009.
[30] L. Singh, S. Tripathi, and H. Arora, "Time Optimization for Traffic Signal Control Using Genetic Algorithm", International Journal of Recent Trends in Engineering, 2 (2), (November 2009), 4-6.
[31] C. Colombaroni, G. Fusco, and A. Gemma, "Optimization of Traffic Signals on Urban Arteries through a Platoon-Based Simulation Model", In Proceedings of the 11th WSEAS International Conference on Automatic Control, Modeling and Simulation, Istanbul, Turkey, May 30 - June 1, 2009.
[32] F. Teklu, A. Sumalee, and D. Watling, A genetic algorithm approach for optimizing traffic control signals considering routing, Comput. Aided Civil and Infrastruct. Eng., vol. 22, no. 1, pp. 31 to 43, 2007.
[33] J. Sanchez, M. Galan, and E. Rubio, Applying a traffic lights evolutionary optimization technique to a real case: las ramblas area in santa cruz de tenerife, IEEE Trans. Evol. Comput., vol. 12, no. 1, pp. 25 to 40, 2008.
[34] Phuong Thi Mai Nguyen, Benjamin N. Passow, Yingjie Yang, Improving Anytime Behavior for Traffic Signal Control Optimization Based on NSGA-II and Local Search, 2016 International Joint Conference on Neural Networks (IJCNN)
[35] J. Chen and L. Xu, Road-junction traffic signal timing optimization by an adaptive particle swarm algorithm, in Proceedings 9th International Conference on Control, Automation, Robotics and Vision, 2006, pp. 1 to 7.

[36] S. Kachroudi and N. Bhouri, A multimodal traffic responsive strategy using particle swarm optimization, in Proceedings of the 12th IFAC Symposium on Transportation Systems, 2009.

[37] J. Garcia-Nieto, E. Alba, and A. Carolina Olivera, Swarm intelligence for traffic light scheduling: Application to real urban areas, Engineering Applications of Artificial Intelligence, vol. 25, no. 2, pp. 274 to 283, 2012.

[38] Garcia-Nieto, J. Ferrer, and E. Alba, Optimising traffic lights with metaheuristics: Reduction of car emissions and consumption, in IJCNN, 2014, pp. 48 to 54.

[39] A. Carolina Olivera, J. Garcia-Nieto, and E. Alba, Reducing vehicle emissions and fuel consumption in the city by using particle swarm optimization, Applied Intelligence, vol. 42, no. 3, pp. 389 to 405, 2015

[40] M A Gokçe, E Oner and G Isik, Traffic signal optimization with Particle Swarm Optimization for signalized roundabouts, Transactions of the Society for Modeling and Simulation International 2015, Vol. 91(5) 456 to 466

[41] Sina Dabiri and Montasir Abbas,"Arterial traffic signal optimization using Particle Swarm Optimization in an Integrated VISSIM-MATLAB simulation environment", 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC) Windsor Oceanico Hotel, Rio de Janeiro, Brazil, November 1-4, 2016.

[42] J. Lee, B. Abdulhai, A. Shalaby, E. Chung,"Real-time optimization for adaptive traffic signal control using genetic algorithms", Journal of Intelligent Transportation Systems Technology, Planning, and Operation, 2005.

[43] PoonamSavsani, VimalSavsani. (2016), Passing vehicle search (PVS): A novel metaheuristic algorithm, Applied Mathematical Modeling Volume 40, Pages 3951-3978

[44] Data provided by "TRANSITEC": independent engineering consultancy firm, specialized in mobility. www.transitec.net